# Resolvent, Continued-Fraction, and Transform Representations of the Fabius–Rvachev System

An audited representation atlas with new frontier results

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**Abstract**

This report audits the LaTeX corpus in the repository subtree `Analysis/FabiusFunction/docs`, including its living expositions, consolidated frontier volume, active draft families, archive lineages, and paper transcriptions [5]. The repository already contains an unusually broad theory of the Fabius function, Rvachev's up-function, their dyadic values, Thue–Morse expansions, Fourier products, moment and cumulant algebra, inverse asymptotics, and spectral decay. The objective here is therefore not to reproduce that material, but to organize it as a representation atlas and then add constructions that are absent from the audited corpus.

The principal new object is the Cauchy–Stieltjes transform

$$R(z) = \int_{-1}^{1} \frac{\mathrm{up}(x)}{z-x}\,\mathrm{d}x.$$

We derive its hyperbolic-product Laplace representation, its one-sided Fourier–Laplace representation, an exact logarithmic fixed-point identity, the functional-differential law

$$R'(z) = 2\big(R(2z+1) - R(2z-1)\big),$$

and an all-orders Thue–Morse derivative tree. Finite dyadic convolution laws yield an explicit $2^N$-term logarithmic formula for $R_N$, whose boundary jump is exactly the finite Thue–Morse spline for the density. This gives a new "logarithm-to-spline" bridge.

We also construct the Jacobi continued fraction of $R$, equivalently the Stieltjes continued fraction of the ordinary even-moment generating function. Its coefficients are positive rational numbers; the first forty are computed exactly. The first coefficients are

$$\beta_1 = \frac{1}{9}, \quad \beta_2 = \frac{32}{225}, \quad \beta_3 = \frac{619}{3675}, \quad \beta_4 = \frac{835232}{4640643}.$$

The Denisov–Rakhmanov theorem proves $\beta_n \to 1/4$, while exact computation through $n = 40$ supports strict monotonicity and a logarithmically corrected convergence law. Further new representations include a trace-class Fredholm determinant for the Fourier image, three equivalent meromorphic logarithmic-derivative expansions, a digamma/Hurwitz- zeta hierarchy, exact one-sided Laplace and Fourier transforms of up and $F$, and pushforward transform identities for the inverse Fabius function. A commented Python program, exact CSV data, numerical checks, and figures accompany the report.

## Contents

# 1 Executive summary and novelty boundary

## 1.1 What was audited

The repository contains several kinds of LaTeX source that must not be conflated:

1. a living primary synthesis of the formalized and analytic theory;
2. a large consolidated non-formalized frontier volume;
3. current draft reports on arithmetic rays, half-integer spectra, self-sampling, inverse asymptotics, Newton/exponent-sequence hierarchies, Fourier decay, and related topics;
4. archived earlier lineages, retained because some contain historically distinctive formulas even when superseded;
5. seven paper-source transcriptions or corrected transcriptions.

The repository's own pinned audit ledger records 67 LaTeX paths at the following commit:

`32d6d36c51d803289e6d6a0dc0c37753766eba47.`

The live tree inspected on 27 August 2026 contains additional current drafts, notably the self-sampling, inverse, and Newton/Rvachev reports. Appendix B records the pinned ledger and the live additions used in this study.

## 1.2 Non-duplication protocol

Every formula in the atlas is tagged conceptually as follows.

| Tag | meaning |
|---|---|
| K | already explicit in the audited repository or classical source; included for orientation |
| D | an immediate transform or recombination of known identities; useful but not claimed as new |
| N | apparently absent from the audited corpus; proved or conjectured here |

The word "new" throughout this report means *new relative to the audited repository.* No claim of global bibliographic priority is made without a separate literature search. In particular, the current frontier report explicitly proposes a Jacobi continued fraction for the moment generating function or Cauchy transform as an open direction. The present report constructs that fraction and develops its consequences.

## 1.3 Main new results

The most substantial additions are:

1. the Cauchy transform $R$ and its moment, Laplace-product, Fourier–Laplace, boundary-value, and logarithmic fixed-point representations;
2. the exact resolvent equation and its all-orders Thue–Morse derivative tree;
3. a finite-level $2^N$-term logarithmic formula converging to $R$, paired by boundary jumps with the finite Thue–Morse spline density;

4. the Jacobi $J$-fraction of $R$, the contracted Stieltjes $S$-fraction of the even moments, exact rational coefficients through order 40, Gaussian quadrature convergents, and the theorem $\beta_n \to 1/4$;
5. a Fredholm determinant realization of the Fourier image and a trace-resolvent interpretation of its logarithmic derivative;
6. cotangent, tangent, valuation-weighted partial-fraction, digamma, and Hurwitz-zeta representations of the logarithmic derivatives of the Fourier image;
7. exact one-sided Laplace and Fourier transforms of up, exact transforms of $F$ on $[0, 1]$, and a master pushforward identity for $F^{-1}$.

## 2 Normalization and structural dictionary

### 2.1 The functions

Let $F : [0, 1] \to [0, 1]$ denote the bounded Fabius function [4, 2], extended by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$. On the unit interval it satisfies

$$F(0) = 0, \qquad F(1) = 1, \qquad F(1-x) = 1 - F(x), \tag{1}$$

$$F'(x) = 2F(2x) \qquad (0 \le x \le \tfrac{1}{2}). \tag{2}$$

The Rvachev function is written

$$\rho(x) := \mathrm{up}(x) = \begin{cases} F(x+1), & -1 \le x \le 0, \\ F(1-x), & 0 \le x \le 1, \\ 0, & |x| \ge 1. \end{cases} \tag{3}$$

Then $\rho$ is even, strictly positive on $(-1, 1)$, normalized by $\int \rho = 1$, flat at $\pm 1$, and satisfies

$$\rho'(x) = 2\big(\rho(2x+1) - \rho(2x-1)\big). \tag{4}$$

### 2.2 Fourier convention and basic transforms

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x}\, \mathrm{d}x, \qquad f(x) = \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i \xi x}\, \mathrm{d}\xi. \tag{5}$$

Set

$$\Phi(\xi) := \widehat{\rho}(\xi), \qquad \mathrm{sinc}_\pi(z) := \frac{\sin(\pi z)}{\pi z}, \quad \mathrm{sinc}_\pi(0) = 1. \tag{6}$$

The classical products are

$$\Phi(z) = \prod_{n=0}^{\infty} \mathrm{sinc}_\pi\Big(\frac{z}{2^n}\Big) = \prod_{m=1}^{\infty} \cos\Big(\frac{\pi z}{2^m}\Big)^m. \tag{7}$$

The scaling law is

$$\Phi(z) = \mathrm{sinc}_\pi(z)\Phi(z/2). \tag{8}$$

### 2.3 The probability model

Let $U_1, U_2, \ldots$ be independent random variables, uniform on $[-1, 1]$, and set

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k. \tag{9}$$

Then $X$ has density $\rho$. Equivalently,

$$X \overset{\mathrm{d}}{=} \frac{U + X'}{2}, \tag{10}$$

where $U \sim \mathrm{Unif}[-1, 1]$ and $X'$ is an independent copy of $X$. The moment generating function is

$$M(t) := \mathbb{E}e^{tX} = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} = \Phi\left(\frac{it}{2\pi}\right). \tag{11}$$

The even moments are

$$\mu_{2n} := \mathbb{E}X^{2n} = \int_{-1}^{1} x^{2n} \rho(x)\, \mathrm{d}x, \qquad \mu_{2n+1} = 0. \tag{12}$$

Finally, if

$$Y = \frac{X + 1}{2}, \tag{13}$$

then $F$ is the distribution function of $Y$, and

$$F'(x) = 2\rho(2x - 1), \qquad 0 < x < 1. \tag{14}$$

## 3 Audited representation atlas

This section compresses the principal representation families already present in the repository and places the new formulas beside them. Proofs of the new entries begin in Section 4.

### 3.1 Real-space and probability representations of $\rho$

**Infinite convolution (K).** Since $2^{-k}U_k$ is uniform on $[-2^{-k}, 2^{-k}]$,

$$\rho = \mathop{\ast}_{k=1}^{\infty} \left(2^{k-1} \mathbf{1}_{[-2^{-k}, 2^{-k}]}\right). \tag{15}$$

This is the real-space counterpart of the sinc product.

**Infinite cube integral (D).** For every bounded continuous test function $g$,

$$\int_{-1}^{1} g(x)\rho(x)\, \mathrm{d}x = \int_{[-1,1]^{\mathbb{N}}} g\left(\sum_{k\ge1} 2^{-k} u_k\right) \prod_{k\ge1} \frac{\mathrm{d}u_k}{2}. \tag{16}$$

Formally, this may be written as a Dirac-delta integral for $\rho(x)$.

**Finite Thue–Morse splines (K).** Put

$$X_N = \sum_{k=1}^{N} 2^{-k} U_k, \qquad a_{N,j} = \frac{2^N - 1 - 2j}{2^N}, \qquad \varepsilon_j = (-1)^{\mathrm{wt}_2(j)}. \tag{17}$$

The density $\rho_N$ of $X_N$ is

$$\rho_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j \, (x + a_{N,j})_+^{N-1}. \tag{18}$$

For each fixed derivative order, $\rho_N \to \rho$ uniformly after $N$ is sufficiently large relative to that order. The repository develops several centered and one-sided variants with explicit error control.

**Fourier inversion and cosine form (K).** Because $\Phi$ is real and even,

$$\rho(x) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i x \xi} \, \mathrm{d}\xi, \tag{19}$$

$$\rho(x) = 2 \int_0^{\infty} \Phi(\xi) \cos(2\pi x \xi) \, \mathrm{d}\xi. \tag{20}$$

The audited corpus also contains Poisson-summation formulas, Legendre expansions, periodized cosine series, dyadic spline limits, and endpoint transseries. They are not rederived here.

### 3.2 Product, zero, gamma, and exponential representations of $\Phi$

**Canonical zero product (K).** Let $\nu_2(n)$ be the exponent of 2 in $n$. The nonzero integer $n$ is a zero of multiplicity $1 + \nu_2(|n|)$, and

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}. \tag{21}$$

**Gamma product (D).** Euler's reflection identity gives

$$\Phi(z) = \prod_{k=0}^{\infty} \frac{1}{\Gamma(1 + z/2^k)\Gamma(1 - z/2^k)}. \tag{22}$$

This is the natural entry point for the Barnes and multiple-gamma compressions studied in current repository drafts.

**Spectral-zeta exponential (K).** For $|z| < 1$,

$$\log \Phi(z) = -\sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(1 - 2^{-2m})} z^{2m}. \tag{23}$$

The coefficient sequence is the spectral zeta function

$$Z_\Phi(s) = \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}. \tag{24}$$

**Fredholm determinant (N).** Section 9 upgrades (21) to

$$\Phi(z) = \det(I - z^2\mathcal{K}), \tag{25}$$

for an explicit positive trace-class diagonal operator $\mathcal{K}$. This makes (24) literally the trace zeta function of $\mathcal{K}$.

### 3.3 Moment, cumulant, and Bell-polynomial representations

The moment generating function has the convergent Taylor series

$$M(t) = \sum_{n=0}^{\infty} \mu_{2n} \frac{t^{2n}}{(2n)!}. \tag{26}$$

Its logarithm is

$$\log M(t) = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!(4^m - 1)} t^{2m}, \tag{27}$$

so all moments are complete Bell polynomials in explicit Bernoulli/Mersenne cumulants. These formulas, together with reciprocal Bell expansions for $1/\Phi$, are developed extensively in the repository.

### 3.4 Transform representations of $F$ and $F^{-1}$

The CDF inversion formula gives

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \Phi(\xi) \frac{\sin(2\pi(2x-1)\xi)}{\xi} \, \mathrm{d}\xi, \qquad 0 < x < 1. \tag{28}$$

Section 7 adds closed Laplace and Fourier transforms of $F$ as a function, as well as transforms of its derivative measure and its inverse.

### 3.5 A compact atlas table

**Table 1:** Representation atlas. Tags refer to the audited-corpus novelty boundary.

| Object | Representation family | Tag | Location |
|---|---|---|---|
| $\rho$ | fold of $F$, compact-support DDE, infinite convolution, random dyadic sum | K | §2–3 |
| $\rho$ | finite Thue–Morse truncated-power splines | K | (18) |
| $\rho$ | Fourier inversion, cosine integral, Poisson and Legendre series | K | (19) |
| $\rho$ | boundary jump of a finite logarithmic resolvent sum | N | Thm. 6.1 |
| $\Phi$ | sinc product and weighted cosine product | K | (7) |
| $\Phi$ | canonical integer-zero product | K | (21) |
| $\Phi$ | gamma/Barnes products | K/D | (22) |
| $\Phi$ | zeta-exponential series | K | (23) |
| $\Phi$ | trace-class Fredholm determinant | N | Thm. 9.2 |
| $\Phi'/\Phi$ | cotangent dyadic sum | N | (60) |
| $\Phi'/\Phi$ | weighted tangent dyadic sum | N | (61) |
| $\Phi'/\Phi$ | valuation-weighted partial fractions | N | (62) |
| $(\log \Phi)^{(r)}$ | digamma and Hurwitz-zeta hierarchy | N | Thm. 8.2 |
| $M$ | hyperbolic-sinc product, moment series, Bell cumulants | K | (11)–(27) |
| $M$ | positive Fredholm determinant | N | (71) |

| Object | Representation family | Tag | Location |
|---|---|---|---|
| $R$ | moment Laurent series | N | (31) |
| $R$ | Laplace transform of the hyperbolic product | N | (32) |
| $R$ | one-sided Fourier–Laplace integral | N | (33) |
| $R$ | logarithmic fixed-point expectation | N | (36) |
| $R$ | dyadic functional-differential equation | N | (37) |
| $R^{(n)}$ | finite Thue–Morse affine orbit of $R$ | N | (38) |
| $R$ | finite $2^N$-logarithm limit | N | (41) |
| $R$ | Jacobi continued fraction | N | (82) |
| $R$ | Gaussian-quadrature partial fractions | N | (91) |
| $\sum \mu_{2n} t^n$ | Stieltjes continued fraction and product integral | N | (84) |
| $F$ | CDF sine-product integral | D | (28) |
| $F$ | exact Laplace transform on $[0, 1]$ | N | (45) |
| $F$ | exact Fourier transform on $[0, 1]$ | N | (46) |
| $\rho\vert_{[0,1]}$ | exact one-sided Laplace/Fourier/sine transforms | N | Thm. 7.2 |
| $F^{-1}$ | pushforward expectation, Laplace, Fourier, and Cauchy transforms | N | Thm. 7.3 |

## 4 The Cauchy–Stieltjes transform

**Definition 4.1.** For $z \in \mathbb{C} \setminus [-1, 1]$, define

$$R(z) = \int_{-1}^{1} \frac{\rho(x)}{z - x} \, \mathrm{d}x = \mathbb{E}\frac{1}{z - X}. \tag{29}$$

The transform $R$ is odd because $\rho$ is even. It is an anti-Herglotz function: when $\operatorname{Im} z > 0$,

$$\operatorname{Im} R(z) = -\operatorname{Im} z \int_{-1}^{1} \frac{\rho(x)}{|z - x|^2} \, \mathrm{d}x < 0. \tag{30}$$

In particular, $R$ has no zeros off the cut. For real $z > 1$, $R(z) > 0$; for $z < -1$, $R(z) < 0$.

**Theorem 4.2** (Four complementary representations of $R$)**.** *The following formulas hold.*

*(i) For* $|z| > 1$,

$$R(z) = \sum_{n=0}^{\infty} \frac{\mu_{2n}}{z^{2n+1}}. \tag{31}$$

*(ii) For* $\operatorname{Re} z > 1$,

$$R(z) = \int_{0}^{\infty} e^{-zt} \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} \, \mathrm{d}t. \tag{32}$$

*(iii) For* $\operatorname{Im} z > 0$,

$$R(z) = -2\pi i \int_{0}^{\infty} e^{2\pi i z\xi} \Phi(\xi) \, \mathrm{d}\xi. \tag{33}$$

*For* $\operatorname{Im} z < 0$, *take the conjugate counterpart.*

*(iv) For* $-1 < x < 1$, *the non-tangential boundary values satisfy*

$$\rho(x) = -\frac{1}{\pi} \lim_{y \downarrow 0} \operatorname{Im} R(x + iy), \tag{34}$$

$$\mathrm{PV} \int_{-1}^{1} \frac{\rho(t)}{x - t} \, \mathrm{d}t = \lim_{y \downarrow 0} \operatorname{Re} R(x + iy) = 2\pi \int_{0}^{\infty} \Phi(\xi) \sin(2\pi x\xi) \, \mathrm{d}\xi. \tag{35}$$

*Proof.* For $|z| > 1$, expand $(z - X)^{-1} = z^{-1} \sum_{m \geq 0} (X/z)^m$, use $|X| \leq 1$, and then symmetry. This gives (31).

For $\operatorname{Re} z > 1$,

$$\frac{1}{z - X} = \int_0^\infty e^{-(z-X)t} \, \mathrm{d}t.$$

Fubini's theorem is justified because $X \leq 1$. Taking expectation and using (11) yields (32).

For $\operatorname{Im} z > 0$,

$$\frac{1}{z - x} = -2\pi i \int_0^\infty e^{2\pi i (z-x)\xi} \, \mathrm{d}\xi.$$

The exponential damping from $\operatorname{Im} z > 0$, together with the rapid decay of $\Phi$, permits Fubini and gives (33). Finally, Plemelj's boundary formula gives (34); taking real parts of (33) gives (35). □

*Remark* 4.3. The four formulas emphasize different structures. The Laurent series exposes exact rational moments; the Laplace integral exposes the hyperbolic product; the Fourier–Laplace integral exposes spectral decay; and the boundary values recover both $\rho$ and its Hilbert transform. Thus $R$ is a single analytic object joining the moment, product, and inversion theories.

# 5 A dyadic functional equation for the resolvent

## 5.1 Logarithmic fixed-point identity

**Theorem 5.1** (Logarithmic fixed point)**.** *For real $z > 1$,*

$$R(z) = \int_{-1}^{1} \rho(y) \log\left(\frac{2z + 1 - y}{2z - 1 - y}\right) \, \mathrm{d}y. \tag{36}$$

*The identity extends by analytic continuation to any simply connected subdomain of $\mathbb{C} \setminus [-1, 1]$, with the logarithm chosen by continuation from large positive $z$.*

*Proof.* Use the fixed point (10) and condition on $X'$:

$$\begin{aligned} R(z) &= \mathbb{E}\left[\int_{-1/2}^{1/2} \frac{\mathrm{d}u}{z - u - X'/2}\right] \\ &= \mathbb{E} \log\left(\frac{z + 1/2 - X'/2}{z - 1/2 - X'/2}\right), \end{aligned}$$

which is (36) after multiplying numerator and denominator by 2. □

## 5.2 Functional-differential equation

**Theorem 5.2** (Resolvent equation)**.** *For every $z \in \mathbb{C} \setminus [-1, 1]$,*

$$\boxed{R'(z) = 2\big(R(2z + 1) - R(2z - 1)\big).} \tag{37}$$

*Proof.* Differentiate (36) first for $z > 1$:

$$\begin{aligned} R'(z) &= \mathbb{E}\left(\frac{2}{2z + 1 - X'} - \frac{2}{2z - 1 - X'}\right) \\ &= 2\big(R(2z + 1) - R(2z - 1)\big). \end{aligned}$$

Both sides are holomorphic on $\mathbb{C} \setminus [-1, 1]$, so the identity theorem gives the stated continuation. □

This is the transform-side analogue of the Rvachev equation (4). It is not merely a Fourier rephrasing: it acts on the spectral parameter of a Cauchy transform and therefore interacts directly with moments, Padé approximation, and orthogonal polynomials.

### 5.3 All-order Thue–Morse derivative tree

**Theorem 5.3** (Thue–Morse derivative orbit)**.** *For every integer $n \geq 0$,*

$$\boxed{R^{(n)}(z) = 2^{n(n+1)/2} \sum_{j=0}^{2^n-1} \varepsilon_j \, R\big(2^n z + 2^n - 1 - 2j\big).} \tag{38}$$

*Consequently,*

$$\int_{-1}^{1} \frac{\rho(x)}{(z-x)^{n+1}} \, \mathrm{d}x = \frac{(-1)^n 2^{n(n+1)/2}}{n!} \sum_{j=0}^{2^n-1} \varepsilon_j R\big(2^n z + 2^n - 1 - 2j\big). \tag{39}$$

*Proof.* The case $n = 0$ is trivial and $n = 1$ is (37). Assume the formula at order $n$. Differentiating contributes a factor $2^n$, and then applying (37) contributes another factor 2. The exponent therefore changes by $n+1$, from $n(n+1)/2$ to $(n+1)(n+2)/2$.

For the shifts, write $a_{n,j} = 2^n - 1 - 2j$. The two children are

$$2a_{n,j} + 1 = 2^{n+1} - 1 - 2(2j), \qquad 2a_{n,j} - 1 = 2^{n+1} - 1 - 2(2j+1).$$

The signs obey $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$, exactly matching the subtraction in (37). This proves (38). Formula (39) follows from

$$R^{(n)}(z) = (-1)^n n! \int_{-1}^{1} \frac{\rho(x)}{(z-x)^{n+1}} \, \mathrm{d}x.$$

□

Taking boundary imaginary parts in (38) recovers the familiar iterated derivative formula for $\rho$. The new point is that the same finite Thue–Morse orbit controls every higher Cauchy kernel, not only the density derivatives.

### 5.4 Uniqueness in the Stieltjes class

**Proposition 5.4.** *Among Cauchy transforms of probability measures supported on $[-1,1]$, $R$ is the unique solution of (37) satisfying*

$$R(z) = \frac{1}{z} + \mathcal{O}(z^{-3}) \qquad (z \to \infty). \tag{40}$$

*Proof.* Let $G(z) = \sum_{r \geq 0} m_r z^{-r-1}$ be the Laurent expansion at infinity of any such Cauchy transform. The normalization gives $m_0 = 1$ and $m_1 = 0$. Substitute this expansion into (37). In the coefficient of $z^{-r-2}$, the contribution containing $m_r$ is

$$-(r+1)m_r + \frac{r+1}{2^r} m_r = -(r+1)(1 - 2^{-r}) m_r,$$

while every other contribution contains only $m_0, \ldots, m_{r-2}$. Since the displayed factor is nonzero for $r \geq 1$, the differential equation determines all moments recursively. The transform $R$ is one solution, so $G$ and $R$ have identical moments. Probability measures on the compact interval $[-1,1]$ are determinate by their moments (equivalently, polynomials are dense in $C[-1,1]$); hence the two measures and their Cauchy transforms coincide. □

## 6 Finite logarithmic resolvents and the spline boundary jump

The finite random sum $X_N$ has two exact representations: its density is a Thue–Morse spline, while its Cauchy transform is a Thue–Morse sum of logarithms. The second formula appears to be absent from the audited corpus.

**Theorem 6.1** (Finite logarithmic formula)**.** *Let*

$$R_N(z) = \mathbb{E}\frac{1}{z - X_N}, \qquad X_N = \sum_{k=1}^{N} 2^{-k} U_k.$$

*For $z \notin [-1 + 2^{-N}, 1 - 2^{-N}]$, with logarithms continued consistently from $z = +\infty$,*

$$\boxed{R_N(z) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N - 1} \varepsilon_j (z + a_{N,j})^{N-1} \operatorname{Log}(z + a_{N,j}).} \tag{41}$$

*Moreover, $R_N \to R$ locally uniformly on $\mathbb{C} \setminus [-1, 1]$, so*

$$R(z) = \lim_{N\to\infty} \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N - 1} \varepsilon_j (z + a_{N,j})^{N-1} \operatorname{Log}(z + a_{N,j}). \tag{42}$$

*Finally, the upper boundary jump of* (41) *is exactly the spline* (18)*:*

$$\rho_N(x) = -\frac{1}{\pi} \operatorname{Im} R_N(x + i0). \tag{43}$$

*Proof.* For $a > 0$, convolution with the uniform density on $[-a, a]$ sends a Cauchy transform $G'(z)$ to

$$\frac{1}{2a} \int_{-a}^{a} G'(z - u)\, \mathrm{d}u = \frac{G(z + a) - G(z - a)}{2a}. \tag{44}$$

Start from the point mass at zero, whose Cauchy transform is $1/z$. An $N$-fold antiderivative is

$$H_N(z) = \frac{z^{N-1} \operatorname{Log} z}{(N-1)!}; \qquad H_N^{(N)}(z) = \frac{1}{z}.$$

Applying (44) successively with $a_k = 2^{-k}$ gives

$$R_N(z) = \frac{1}{\prod_{k=1}^{N} 2a_k} \sum_{\sigma \in \{\pm 1\}^N} \left( \prod_{k=1}^{N} \sigma_k \right) H_N\left( z + \sum_{k=1}^{N} \sigma_k a_k \right).$$

Since

$$\frac{1}{\prod_{k=1}^{N} 2a_k} = 2^{N(N-1)/2},$$

and binary subsets identify the shifts and signs with $a_{N,j}$ and $\varepsilon_j$, this is (41).

The residual tail $X - X_N$ has support in $[-2^{-N}, 2^{-N}]$, so $X_N \to X$ almost surely and uniformly in support. The kernels $(z - x)^{-1}$ are uniformly continuous on compact subsets away from $[-1, 1]$; hence $R_N \to R$ locally uniformly.

For the boundary value, $\operatorname{Im} \operatorname{Log}(x + a + i0) = \pi$ when $x + a < 0$ and zero when $x + a > 0$. The alternating finite difference annihilates every polynomial of degree less than $N$, so the sum of the full powers $(x + a_{N,j})^{N-1}$ is zero. Replacing the negative-part sum by minus the positive-part sum yields exactly (18). This proves (43). ☐

*Remark* 6.2 (Two complementary approximation geometries). The finite-convolution approximant $R_N$ is a sum of logarithms and preserves the dyadic construction. The Jacobi approximants in Section 10 are rational functions and preserve moments. Both converge locally uniformly off the support, but their poles, branch cuts, and error mechanisms are completely different. This duality suggests hybrid algorithms: use finite convolution to resolve endpoint geometry and a short Jacobi fraction to compress the remaining moment error.

# 7 Exact transforms of $F$, one-sided up, and $F^{-1}$

## 7.1 The Laplace and Fourier transforms of $F$

**Theorem 7.1** (Transform of the Fabius function)**.** *For complex $s$, with the value at $s=0$ understood by continuity,*

$$\boxed{\int_0^1 e^{-sx}F(x)\,\mathrm{d}x = \frac{e^{-s/2}M(s/2)-e^{-s}}{s}.} \tag{45}$$

*Equivalently, for real or complex $\xi$,*

$$\boxed{\int_0^1 F(x)e^{-2\pi i\xi x}\,\mathrm{d}x = \frac{e^{-\pi i\xi}\Phi(\xi/2)-e^{-2\pi i\xi}}{2\pi i\xi}.} \tag{46}$$

*At $\xi=0$, the value is $1/2$.*

*Proof.* The distributional Laplace transform of $Y$ is

$$\int_0^1 e^{-sx}\,\mathrm{d}F(x) = \mathbb{E}e^{-sY} = e^{-s/2}M(s/2). \tag{47}$$

Integration by parts gives

$$\int_0^1 e^{-sx}\,\mathrm{d}F(x) = e^{-s} + s\int_0^1 e^{-sx}F(x)\,\mathrm{d}x,$$

which is (45). Substitute $s=2\pi i\xi$ and use $M(\pi i\xi)=\Phi(\xi/2)$. □

Separating real and imaginary parts gives the explicit sine and cosine transforms

$$\int_0^1 F(x)\cos(2\pi\xi x)\,\mathrm{d}x = \frac{\sin(2\pi\xi)-\Phi(\xi/2)\sin(\pi\xi)}{2\pi\xi}, \tag{48}$$

$$\int_0^1 F(x)\sin(2\pi\xi x)\,\mathrm{d}x = \frac{\Phi(\xi/2)\cos(\pi\xi)-\cos(2\pi\xi)}{2\pi\xi}. \tag{49}$$

## 7.2 One-sided transforms of the up-function

On $[0,1]$, $\rho(x)=F(1-x)=1-F(x)$. Therefore the previous theorem immediately yields the following closed forms.

**Theorem 7.2** (One-sided transform package)**.** *For complex $s$, with the removable value at zero,*

$$\boxed{\int_0^1 e^{-sx}\rho(x)\,\mathrm{d}x = \frac{1-e^{-s/2}M(s/2)}{s}.} \tag{50}$$

*Consequently,*

$$\boxed{\int_0^1 e^{-2\pi i\xi x}\rho(x)\,\mathrm{d}x = \frac{1-e^{-\pi i\xi}\Phi(\xi/2)}{2\pi i\xi}.} \tag{51}$$

*The half-line cosine and sine transforms are*

$$\int_0^1 \rho(x)\cos(2\pi\xi x)\,\mathrm{d}x = \frac{1}{2}\Phi(\xi), \tag{52}$$

$$\int_0^1 \rho(x)\sin(2\pi\xi x)\,\mathrm{d}x = \frac{1-\Phi(\xi/2)\cos(\pi\xi)}{2\pi\xi}. \tag{53}$$

The sum of (50) at $s$ and $-s$ recovers $M(s)$ precisely because

$$M(s) = \frac{2\sinh(s/2)}{s}M(s/2), \tag{54}$$

which is the hyperbolic continuation of (8).

### 7.3 The inverse Fabius function as a transport map

Let $I = F^{-1}$ on $[0,1]$. Since $I$ pushes Lebesgue measure forward to the law of $Y$, every integral of a composition with $I$ reduces to the up-law.

**Theorem 7.3** (Inverse pushforward master identity). *For every Borel function $g$ for which one side is integrable,*

$$\boxed{\int_0^1 g(I(u))\,\mathrm{d}u = \int_0^1 g(x)\,\mathrm{d}F(x) = \int_{-1}^1 g\left(\frac{x+1}{2}\right)\rho(x)\,\mathrm{d}x.} \tag{55}$$

*In particular,*

$$\int_0^1 e^{-sI(u)}\,\mathrm{d}u = e^{-s/2}M(s/2), \tag{56}$$

$$\int_0^1 e^{-2\pi i\xi I(u)}\,\mathrm{d}u = e^{-\pi i\xi}\Phi(\xi/2), \tag{57}$$

$$\int_0^1 \frac{\mathrm{d}u}{z-I(u)} = 2R(2z-1), \qquad z\in\mathbb{C}\setminus[0,1], \tag{58}$$

$$\int_0^1 I(u)^n\,\mathrm{d}u = 2^{-n}\sum_{r=0}^{\lfloor n/2\rfloor}\binom{n}{2r}\mu_{2r}. \tag{59}$$

*Proof.* If $U$ is uniform on $[0,1]$, then $I(U)$ has CDF $F$, hence the same law as $Y$. This proves (55). The remaining formulas follow by taking $g(x) = e^{-sx}$, $e^{-2\pi i\xi x}$, $(z-x)^{-1}$, and $x^n$, respectively, and then using (13) and symmetry of $X$. □

*Remark* 7.4. Formula (58) transports the entire Jacobi-fraction theory of $R$ to the inverse Fabius function. Thus a function whose pointwise inverse asymptotics involve Lambert $W$ nevertheless has a globally simple Cauchy transform when sampled with uniform measure in its range variable.

# 8 Logarithmic derivatives of the Fourier image

## 8.1 Three meromorphic expansions

**Theorem 8.1** (Cotangent, tangent, and zero-set formulas)**.** *Away from the integer zero set of* $\Phi$*,*

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} \left[ \frac{\pi}{2^k} \cot\left(\frac{\pi z}{2^k}\right) - \frac{1}{z} \right], \tag{60}$$

$$= -\pi \sum_{m=1}^{\infty} \frac{m}{2^m} \tan\left(\frac{\pi z}{2^m}\right), \tag{61}$$

$$= -2z \sum_{n=1}^{\infty} \frac{1+\nu_2(n)}{n^2 - z^2}. \tag{62}$$

*All three series converge locally normally on compact subsets avoiding the poles.*

*Proof.* Differentiate the logarithm of each of the three products (7) and (21). For large $k$, the bracket in (60) and the term in (61) are $\mathcal{O}(4^{-k})$ uniformly on compact sets. The terms in (62) are bounded by a constant times $(1 + \nu_2(n))/n^2$, which is summable by (24) at $s = 2$. □

The three forms have different numerical regimes. The tangent sum is usually the fastest away from its poles; the cotangent sum tracks the original sinc factors; and the zero-set sum exposes multiplicities and supports rigorous rational tail bounds.

## 8.2 Digamma and Hurwitz-zeta hierarchy

The valuation weight has the divisor decomposition

$$1 + \nu_2(n) = \sum_{k=0}^{\infty} \mathbf{1}_{2^k|n}. \tag{63}$$

This turns the zero sum into a dyadic superposition of classical special functions.

**Theorem 8.2** (Digamma and Hurwitz-zeta representations)**.** *Let* $\psi = \Gamma'/\Gamma$*. Then*

$$\boxed{\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} 2^{-k} \left[ \psi\left(1 - \frac{z}{2^k}\right) - \psi\left(1 + \frac{z}{2^k}\right) \right].} \tag{64}$$

*For every integer* $r \geq 2$*,*

$$\boxed{\frac{\mathrm{d}^r}{\mathrm{d}z^r} \log \Phi(z) = (r-1)! \sum_{k=0}^{\infty} 2^{-kr} \left[ -\zeta\left(r, 1 - \frac{z}{2^k}\right) + (-1)^{r-1} \zeta\left(r, 1 + \frac{z}{2^k}\right) \right].} \tag{65}$$

*Proof.* From (21), for $r \geq 1$, paired symmetrically when $r = 1$,

$$(\log \Phi)^{(r)}(z) = (r-1)! \sum_{n \geq 1} (1 + \nu_2(n)) \left[ -\frac{1}{(n-z)^r} + \frac{(-1)^{r-1}}{(n+z)^r} \right]. \tag{66}$$

Insert (63), write $n = 2^k m$, and sum over $m \geq 1$. For $r \geq 2$, the inner sums are Hurwitz zeta functions. For $r = 1$, the convergent paired sum is the standard digamma difference. □

**Corollary 8.3** (Local jet data at nonzero zeros). *Let $N \neq 0$ be an integer and $d_N = 1 + \nu_2(|N|)$. Then*

$$\frac{\Phi'(z)}{\Phi(z)} = \frac{d_N}{z - N} + H_N(z), \tag{67}$$

*where $H_N$ is holomorphic near $N$, and every derivative of $H_N$ can be written as an absolutely convergent valuation-weighted Hurwitz-zeta sum. Thus the complete local Taylor jet of $\Phi(z)/(z - N)^{d_N}$ is algorithmically explicit.*

## 9 A Fredholm determinant for the Fourier product

The canonical product can be realized as an ordinary Fredholm determinant, not merely as a formal spectral product.

**Definition 9.1.** Let

$$\mathcal{I} = \{(n, r) : n \geq 1,\ 1 \leq r \leq 1 + \nu_2(n)\}, \tag{68}$$

and define the positive diagonal operator $\mathcal{K}$ on $\ell^2(\mathcal{I})$ by

$$\mathcal{K} e_{n,r} = \frac{1}{n^2} e_{n,r}. \tag{69}$$

**Theorem 9.2** (Fredholm determinant representation). *The operator $\mathcal{K}$ is trace class and*

$$\boxed{\Phi(z) = \det(I - z^2\mathcal{K})}, \tag{70}$$

$$\boxed{M(t) = \det\left(I + \frac{t^2}{4\pi^2}\mathcal{K}\right)}. \tag{71}$$

*Furthermore,*

$$\operatorname{tr}(\mathcal{K}^m) = \frac{\zeta(2m)}{1 - 2^{-2m}}, \tag{72}$$

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \operatorname{tr}\big(\mathcal{K}(I - z^2\mathcal{K})^{-1}\big). \tag{73}$$

*Proof.* By (24),

$$\operatorname{tr}\mathcal{K} = \sum_{n \geq 1} \frac{1 + \nu_2(n)}{n^2} = \frac{\zeta(2)}{1 - 2^{-2}} < \infty,$$

so $\mathcal{K}$ is trace class. Its Fredholm determinant is the product over its eigenvalues, with multiplicity:

$$\det(I - z^2\mathcal{K}) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)} = \Phi(z).$$

Substituting $z = it/(2\pi)$ gives (71). Formula (72) is again (24), and differentiating the Fredholm logarithm gives (73). □

*Remark* 9.3. This representation converts several scalar identities into operator identities. For example, the zeta expansion (23) is the standard trace expansion

$$\log\det(I - z^2\mathcal{K}) = -\sum_{m \geq 1} \frac{z^{2m}}{m} \operatorname{tr}(\mathcal{K}^m).$$

It also suggests perturbation theory for generalized exponent sequences: changing the multiplicity sequence changes only the diagonal spectral multiplicities.

# 10 Jacobi continued fractions and exact rational coefficients

## 10.1 Exact moment recurrence

**Proposition 10.1** (Moment recurrence). *The even moments satisfy $\mu_0 = 1$ and, for $n \geq 1$,*

$$\boxed{(4^n - 1)\mu_{2n} = \sum_{j=1}^{n} \binom{2n}{2j} \frac{\mu_{2n-2j}}{2j+1}.} \tag{74}$$

*In particular, every $\mu_{2n}$ is rational.*

*Proof.* Raise (10) to the $2n$-th power and take expectations:

$$4^n \mu_{2n} = \sum_{j=0}^{n} \binom{2n}{2j} \mathbb{E}U^{2j}\, \mu_{2n-2j}.$$

Since $\mathbb{E}U^{2j} = 1/(2j+1)$, the $j = 0$ term is $\mu_{2n}$. Move it to the left. □

## 10.2 Monic orthogonal polynomials

Let $P_n$ be the monic orthogonal polynomials for the probability measure $\rho(x)\,\mathrm{d}x$, and let

$$h_n = \int_{-1}^{1} P_n(x)^2 \rho(x)\ \mathrm{d}x. \tag{75}$$

Evenness forces the diagonal Jacobi parameters to vanish, so

$$P_0(x) = 1,\ P_1(x) = x, \qquad P_{n+1}(x) = xP_n(x) - \beta_n P_{n-1}(x), \qquad \beta_n = \frac{h_n}{h_{n-1}} > 0. \tag{76}$$

The first polynomials are

$$P_0(x) = 1, \tag{77}$$

$$P_1(x) = x, \tag{78}$$

$$P_2(x) = x^2 - \frac{1}{9}, \tag{79}$$

$$P_3(x) = x^3 - \frac{19}{75}x, \tag{80}$$

$$P_4(x) = x^4 - \frac{62}{147}x^2 + \frac{619}{33075}. \tag{81}$$

## 10.3 The Jacobi and Stieltjes fractions

**Theorem 10.2** (Jacobi continued fraction). *The Cauchy transform has the locally uniform continued fraction*

$$\boxed{R(z) = \cfrac{1}{z - \cfrac{\beta_1}{z - \cfrac{\beta_2}{z - \cfrac{\beta_3}{z - \ddots}}}}.} \tag{82}$$

*The coefficients are positive rational numbers.*

*Proof.* Positivity follows from $\beta_n = h_n/h_{n-1}$, and rationality follows because Gram–Schmidt applied to the rational moment functional (74) uses only rational operations. Markov's theorem for compactly supported positive measures identifies the continued fraction generated by (76) with the Cauchy transform and gives local uniform convergence off the support [7, 6]. □

Define the ordinary even-moment generating function

$$\mathcal{O}(t) = \sum_{n=0}^{\infty} \mu_{2n} t^n = \int_{-1}^{1} \frac{\rho(x)}{1 - tx^2}\, \mathrm{d}x, \qquad |t| < 1. \tag{83}$$

Substituting $t = z^{-2}$ into (82) gives the contracted fraction.

**Corollary 10.3** (Stieltjes fraction and product integral)**.** *For* $|t| < 1$,

$$\boxed{\mathcal{O}(t) = \cfrac{1}{1 - \cfrac{\beta_1 t}{1 - \cfrac{\beta_2 t}{1 - \cfrac{\beta_3 t}{1 - \ddots}}}}} \tag{84}$$

$$\mathcal{O}(t) = \int_0^{\infty} e^{-u} \prod_{k=1}^{\infty} \frac{\sinh(\sqrt{t}\, u/2^k)}{\sqrt{t}\, u/2^k}\, \mathrm{d}u. \tag{85}$$

*Proof.* The continued fraction follows from $\mathcal{O}(z^{-2}) = zR(z)$. For the integral, use

$$\frac{1}{1 - tx^2} = \int_0^{\infty} e^{-u} \cosh(\sqrt{t}\, xu)\, \mathrm{d}u$$

and then average over $X$, obtaining $M(\sqrt{t}\, u)$. □

For $s \geq 0$, $\mathcal{O}(-s)$ is completely monotone:

$$(-1)^m \frac{\mathrm{d}^m}{\mathrm{d}s^m} \mathcal{O}(-s) = m! \int_{-1}^{1} \frac{x^{2m}\rho(x)}{(1 + sx^2)^{m+1}}\, \mathrm{d}x > 0. \tag{86}$$

Thus the even moments form a Stieltjes moment sequence, and the fraction (84) belongs to the classical positive Stieltjes class.

### 10.4 Hankel determinants

Let

$$\Delta_n = \det[\mu_{i+j}]_{i,j=0}^{n-1}, \qquad \Delta_0 = 1. \tag{87}$$

Then

$$h_n = \frac{\Delta_{n+1}}{\Delta_n}, \qquad \boxed{\beta_n = \frac{\Delta_{n+1}\Delta_{n-1}}{\Delta_n^2}}. \tag{88}$$

Because odd moments vanish, each $\Delta_n$ factors after a parity permutation into an even-even and odd-odd moment determinant. This parity factorization is useful for exact arithmetic and should be exploited in future Lean formalization.

### 10.5 Exact coefficients and numerical evidence

The accompanying Python program computes the moments by (74) and then uses the exact symmetric Stieltjes procedure (76). Selected results are:

| $n$ | $\mu_{2n}$ | decimal |
|---|---|---|
| 0 | $1$ | 1 |
| 1 | $\frac{1}{9}$ | 0.111111111111 |
| 2 | $\frac{19}{675}$ | 0.0281481481481 |
| 3 | $\frac{583}{59535}$ | 0.00979255899891 |
| 4 | $\frac{132809}{32531625}$ | 0.00408245822334 |
| 5 | $\frac{46840699}{24405225075}$ | 0.00191928977733 |
| 6 | $\frac{4068990560161}{4133856862760625}$ | 0.000984308527181 |
| 7 | $\frac{1204567303451311}{2232691548877164375}$ | 0.000539513532022 |
| 8 | $\frac{4146897304424408411}{13301767332333178846875}$ | 0.00031175536309 |
| 9 | $\frac{1881436000669580752786879 3}{}$ | |

| $n$ | $\beta_n$ (exact when compact) | decimal |
|---|---|---|
| 1 | $\frac{1}{9}$ | 0.111111111111 |
| 2 | $\frac{32}{225}$ | 0.142222222222 |
| 3 | $\frac{619}{3675}$ | 0.168435374150 |
| 4 | $\frac{835232}{4640643}$ | 0.179981955087 |
| 5 | $\frac{1789043898625}{9272274723657}$ | 0.192945523288 |
| 6 | see data file | 0.199759514028 |
| 7 | see data file | 0.205964295192 |
| 8 | see data file | 0.211147954951 |
| 9 | see data file | 0.215261988744 |
| 10 | see data file | 0.218406233337 |
| 11 | see data file | 0.221259971300 |
| 12 | see data file | 0.223692820062 |

The full numerators and denominators through $n = 40$ are in `up_jacobi_coefficients.csv`. Their size grows rapidly: by moderate order the reduced integers contain thousands of decimal digits. This arithmetic complexity is not a floating-point artifact.

### 10.6 The limiting coefficient

**Theorem 10.4** (Nevai-class limit)**.** *The Jacobi coefficients satisfy*

$$\boxed{\lim_{n\to\infty} \beta_n = \frac{1}{4}.} \tag{89}$$

*Proof.* The measure $\rho(x)\,\mathrm{d}x$ has essential support $[-1, 1]$, and its density is strictly positive for every $x \in (-1, 1)$. The Denisov–Rakhmanov theorem therefore places its orthonormal Jacobi matrix in the Nevai class of $[-1, 1]$ [3, 6]: the off-diagonal orthonormal coefficients tend to $1/2$, and the diagonal coefficients tend to zero. In the monic recurrence (76), $\beta_n$ is the square of the off-diagonal orthonormal coefficient, hence tends to $1/4$. $\square$

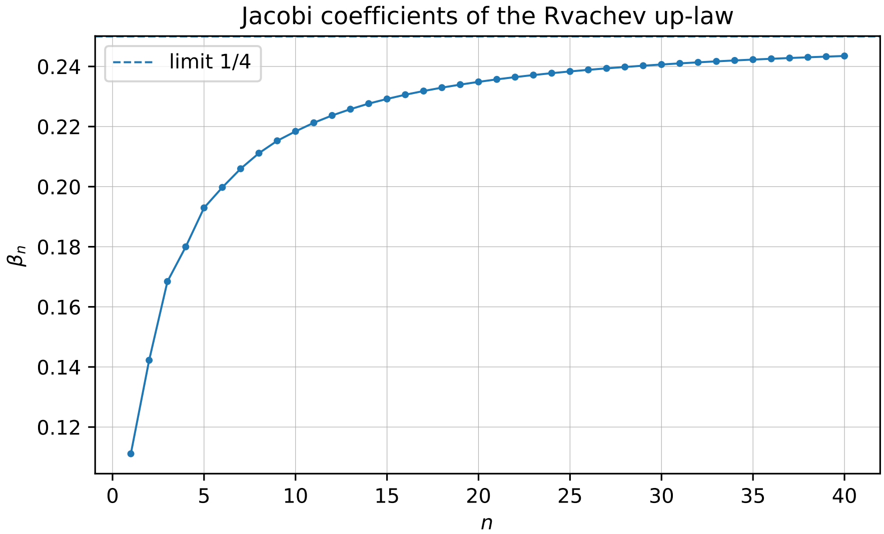


**Figure 1:** Exact Jacobi coefficients converted to decimals. The dashed line is the proved limit $1/4$.

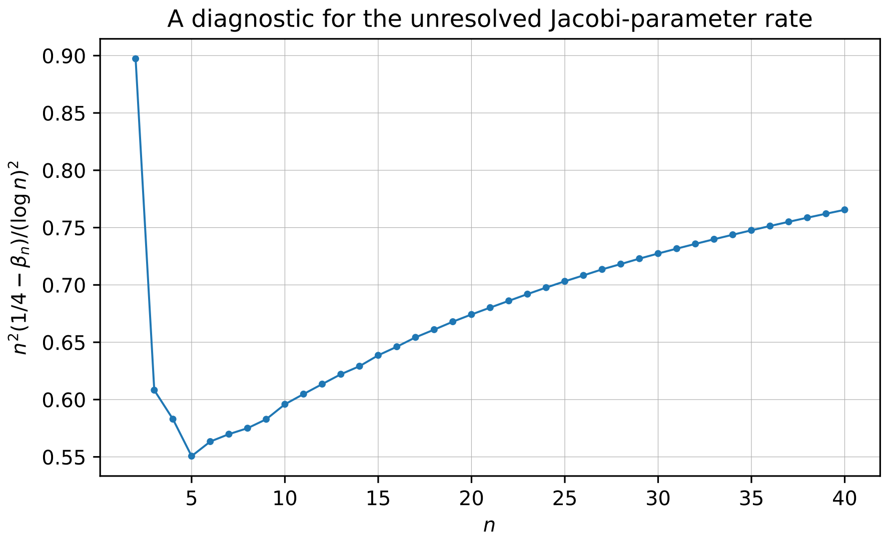


**Figure 2:** The diagnostic quantity $n^2(1/4 - \beta_n)/(\log n)^2$. The range $n \leq 40$ is evidence, not an asymptotic proof.

### 10.7 Padé and Gaussian-quadrature representations

Let $Q_{N-1}$ be the polynomial of the second kind

$$Q_{N-1}(z) = \int_{-1}^{1} \frac{P_N(z) - P_N(x)}{z - x} \rho(x)\, \mathrm{d}x. \tag{90}$$

The $N$-th Jacobi convergent is $Q_{N-1}/P_N$. If $x_{1,N}, \ldots, x_{N,N}$ are the zeros of $P_N$, then

$$\boxed{\frac{Q_{N-1}(z)}{P_N(z)} = \sum_{j=1}^{N} \frac{w_{j,N}}{z - x_{j,N}}, \qquad w_{j,N} > 0, \quad \sum_j w_{j,N} = 1.} \tag{91}$$

This is the Cauchy transform of the $N$-node Gaussian quadrature measure for $\rho$, and it matches moments through degree $2N - 1$. The exact error identity is

$$R(z) - \frac{Q_{N-1}(z)}{P_N(z)} = \frac{1}{P_N(z)^2} \int_{-1}^{1} \frac{P_N(x)^2}{z - x} \rho(x)\, \mathrm{d}x. \tag{92}$$

Hence every Gaussian resolvent approximant is a strict lower bound for $R(z)$ when $z > 1$, and a strict upper bound when $z < -1$.

The first nontrivial approximants are completely explicit:

$$R_1^{\mathrm{G}}(z) = \frac{1}{z}, \tag{93}$$

$$R_2^{\mathrm{G}}(z) = \frac{z}{z^2 - 1/9} = \frac{1}{2}\left(\frac{1}{z - 1/3} + \frac{1}{z + 1/3}\right), \tag{94}$$

$$R_3^{\mathrm{G}}(z) = \frac{32/57}{z} + \frac{25/114}{z - \sqrt{19/75}} + \frac{25/114}{z + \sqrt{19/75}}. \tag{95}$$

Thus the continued fraction gives a hierarchy of positive finite atomic models for the up-law, complementary to the continuous finite-convolution models of Section 5.

## 11 Further combined representations

### 11.1 The Cauchy transform of the Fabius measure

Let

$$S(z) = \int_0^1 \frac{\mathrm{d}F(x)}{z - x} = \mathbb{E}\frac{1}{z - Y}. \tag{96}$$

The affine relation (13) gives

$$\boxed{S(z) = 2R(2z - 1).} \tag{97}$$

Consequently, every representation of $R$ immediately yields one for $S$. For example, for $|2z - 1| > 1$,

$$S(z) = 2\sum_{n=0}^{\infty} \frac{\mu_{2n}}{(2z - 1)^{2n+1}}, \tag{98}$$

and $S$ has a Jacobi fraction obtained by the affine change of variable in (82).

### 11.2 A determinant form of the Fabius Laplace transform

Combining (45) with (71) gives

$$\int_0^1 e^{-sx} F(x)\, \mathrm{d}x = \frac{e^{-s/2} \det\left(I + \frac{s^2}{16\pi^2}\mathcal{K}\right) - e^{-s}}{s}. \tag{99}$$

Similarly,

$$\int_0^1 e^{-sx} \rho(x)\, \mathrm{d}x = \frac{1 - e^{-s/2} \det\left(I + \frac{s^2}{16\pi^2}\mathcal{K}\right)}{s}. \tag{100}$$

These are literal combinations of a compact-support transform with a trace-class spectral determinant.

### 11.3 A resolvent–product–continued-fraction triangle

For $|t| < 1$, the same analytic function $\mathcal{O}(t)$ has three structurally different representations:

$$\boxed{\begin{aligned} \mathcal{O}(t) &= \sum_{n\ge 0} \mu_{2n} t^n \\ &= \int_0^\infty e^{-u} \prod_{k\ge 1} \frac{\sinh(\sqrt{t}\,u/2^k)}{\sqrt{t}\,u/2^k}\, \mathrm{d}u \\ &= \cfrac{1}{1 - \cfrac{\beta_1 t}{1 - \cfrac{\beta_2 t}{1 - \ddots}}}. \end{aligned}} \tag{101}$$

The first line is arithmetic, the second is multiplicative and analytic, and the third is orthogonal-polynomial/geometric. Equality of the three creates a direct bridge among the Bernoulli–Bell moment formulas, the infinite hyperbolic product, and the exact rational Jacobi parameters.

## 12 Numerical experiments and reproducibility

### 12.1 Exact arithmetic

The script `rvachev_frontier_experiments.py` performs the following operations.

1. It computes $\mu_0, \mu_2, \ldots$ exactly using (74).
2. It constructs the monic orthogonal polynomials using the symmetric three-term recurrence and exact rational inner products.
3. It writes all exact $\beta_n$ through a user-selected order to CSV.
4. It checks (37) independently using a long moment expansion.
5. It compares the cotangent, tangent, and zero-set formulas for $\Phi'/\Phi$.
6. It generates Figures 1 and 2.

The default order is 40. On the reference environment the full run takes tens of seconds; the exact integers, rather than floating-point work, dominate at higher order.

### 12.2 Numerical verification of the resolvent equation

At 45-decimal mpmath precision, using moment expansions through orders 160 and 220, the computed values were:

| $z$ | $R'(z)$ | $2(R(2z+1)-R(2z-1))$ | relative discrepancy |
|---|---|---|---|
| 1.25 | $-0.831285009171483062344621$4 | same | $3.6 \times 10^{-46}$ |
| 1.75 | $-0.367918959914653875380379$7 | same | below printed precision |
| 2.50 | $-0.169158976095556300174058$3 | same | $4.4 \times 10^{-47}$ |

At $z = 1.25$, the change between the two truncation lengths was about $3.8 \times 10^{-41}$, so the identity error is below the visible truncation scale.

### 12.3 Numerical comparison of logarithmic derivatives

The cotangent and tangent formulas agreed to roughly 44 decimal places at the tested nonzero points. The zero-set formula, truncated at $10^5$, differed by $5.2 \times 10^{-6}$, $1.5 \times 10^{-5}$, and $2.9 \times 10^{-5}$ at $z = 0.13, 0.37, 0.73$, respectively, consistent with its algebraic tail. The data file `numerical_checks.txt` records all values.

## 13 Conjectures and frontier problems

The following conjectures are deliberately separated from the proved results.

**Conjecture 13.1** (Strict Jacobi monotonicity)**.** The exact Jacobi coefficients are strictly increasing:

$$0 < \beta_1 < \beta_2 < \cdots < \frac{1}{4}. \tag{102}$$

Exact arithmetic verifies the inequalities through $n = 40$. The upper limit is proved by Theorem 10.4, but monotonicity does not follow from the Denisov–Rakhmanov theorem and appears to require a Fabius-specific argument.

**Conjecture 13.2** (Logarithmically corrected recurrence rate)**.** There exists a constant $C_{\mathrm{up}} \in (0, \infty)$ such that

$$\frac{1}{4} - \beta_n \sim C_{\mathrm{up}} \frac{(\log n)^2}{n^2}. \tag{103}$$

The scaled quantity in Figure 2 rises from approximately $0.674$ at $n = 20$ to $0.766$ at $n = 40$. This is too short a range to determine a constant or even to exclude a different slowly varying factor. The conjecture is motivated jointly by the data and by the squared-logarithmic endpoint scale of $\rho$. A Riemann–Hilbert or strong Szegő analysis adapted to the endpoint flatness is the natural next step.

**Conjecture 13.3** (Dyadically regularized Mittag–Leffler expansion)**.** Let $d_n = 1 + \nu_2(|n|)$, and let $\mathrm{PP}_n(z)$ be the principal part of $1/\Phi(z)$ at the nonzero integer $n$. There exists a canonical dyadic grouping and a finite genus of subtraction polynomials such that

$$\frac{1}{\Phi(z)} = E(z) + \sum_{n\in\mathbb{Z}\setminus\{0\}}^{\text{dyadic}} \left(\mathrm{PP}_n(z) - \text{canonical subtraction at } n\right), \tag{104}$$

where $E$ is an explicitly controlled entire function, ideally constant or of very low order.

The local principal parts are computable from Theorem 8.2; the unresolved issue is a global summation scheme compatible with the unbounded dyadic multiplicities.

**Problem 13.4** (Arithmetic of $\beta_n$)**.** Determine sharp denominator-clearing factors for $\beta_n$, $h_n$, and the parity Hankel determinants. The first coefficients display repeated migration of prime factors between consecutive norms, suggesting that factored norm variables may be arithmetically simpler than reduced $\beta_n$ themselves.

**Problem 13.5** (Endpoint-sensitive Gaussian nodes)**.** Determine the asymptotic distance of the extreme zero of $P_n$ from 1, and compare it with the Chebyshev scale $n^{-2}$. The density is positive almost everywhere but flatter than every power at the endpoint, so the leading equilibrium distribution is classical while the first correction should retain the Lambert–$W$/log-square endpoint signature.

**Problem 13.6** (A coupled resolvent hierarchy for exponent sequences)**.** The repository's Newton/Rvachev theory replaces the unit exponent sequence in the sinc product by general integer sequences. Construct the corresponding Cauchy transforms and derive coupled affine functional equations when the exponent sequence is closed under a Pascal or shift operator. The Fredholm model of Section 9 suggests that the Fourier side is easy; the real-space fixed-point law is the main obstacle.

**Problem 13.7** (Fractional resolvent dynamics)**.** Equation (37) gives every integer derivative as a finite Thue–Morse orbit. Seek a fractional-order interpolation

$$D_z^\alpha R(z) = \int_{\Gamma_\alpha} K_\alpha(j) R(2^\alpha z + \text{affine shift}(j)) \, \mathrm{d}j$$

that reduces to (38) for $\alpha \in \mathbb{N}$. A semigroup built from the two affine branches $z \mapsto 2z \pm 1$ may provide the correct operator calculus.

**Problem 13.8** (Formal verification roadmap)**.** A Lean development can be staged as follows:

1. define the Cauchy transform of the canonical up probability measure;
2. prove the logarithmic fixed-point identity first on $(1, \infty)$;
3. differentiate under the integral and formalize (37);
4. prove the Thue–Morse derivative tree by induction on binary children;
5. define rational moments and the monic Stieltjes procedure;
6. construct finite Jacobi convergents and prove the Gaussian error identity;
7. postpone the Denisov–Rakhmanov limit, which depends on substantial external orthogonal-polynomial theory.

# 14 Promising research program

## 14.1 Analytic program

The Cauchy transform should be treated as a primary Fabius–Rvachev object, on the same level as $F$, $\rho$, and $\Phi$. Immediate goals are:

1. derive quantitative bounds for $R$ near the cut from the sharp Fourier decay;
2. study the Hilbert transform (35), including its zeros and endpoint asymptotics;
3. use (37) to continue asymptotic expansions along dyadic affine rays;
4. compare finite-logarithmic and Jacobi rational approximants uniformly near the support.

### 14.2 Orthogonal-polynomial program

The exact $J$-fraction opens a separate research direction:

1. prove or disprove Conjecture 13.1;
2. determine strong asymptotics of $P_n$, $h_n$, and $\beta_n$;
3. construct Fabius-specific Gauss–Kronrod or nested positive rules;
4. investigate whether the dyadic moment recurrence induces a nonlinear recurrence or integrable system for $\beta_n$;
5. study the arithmetic Galois structure of Gaussian nodes and exact weights.

### 14.3 Operator and special-function program

The Fredholm determinant and Hurwitz-zeta hierarchy suggest:

1. regularized determinants and spectral shifts for deformed multiplicity sequences;
2. a Barnes-type compression of the determinant rather than only of the scalar product;
3. exact trace identities for logarithmic derivatives at rational and half-integer arguments;
4. meromorphic continuation of dyadically weighted spectral sums with local factors dictated by $\nu_2(n)$.

### 14.4 Inverse-function program

The pushforward formulas (55)–(58) show that many global integral questions about $F^{-1}$ are easier than pointwise questions. Promising directions include:

1. exact orthogonal expansions of $I(u)$ using its known pushforward moments;
2. entropy and transport functionals of $I$, reduced to integrals against $\rho$;
3. combining the global Cauchy transform with local Lambert–$W$ endpoint inversions;
4. rational and Gaussian quadrature for inverse-function integrals without directly evaluating $I$.

## 15 Conclusion

The audited repository already contains most of the classical and modern representation theory one would first seek: convolution laws, sinc and cosine products, integer-zero products, moment and cumulant series, Thue–Morse splines, q-binomial dyadic values, Poisson and Legendre expansions, inverse asymptotics, and Fourier-decay envelopes. The most productive way to extend that corpus is therefore to introduce a new organizing transform rather than another isolated formula.

The Cauchy transform $R$ accomplishes this. It joins the moment sequence to an analytic function, converts the dyadic law into a functional-differential equation, turns finite convolutions into logarithmic Thue–Morse sums, and generates a Jacobi continued fraction with exact rational coefficients. The Fourier image simultaneously acquires an operator determinant and a special-function logarithmic-derivative calculus. Closed one-sided transforms and inverse pushforward identities complete a network in which real-space, frequency-space, moment, orthogonal-polynomial, and operator representations are mutually convertible.

The proved limit $\beta_n \to 1/4$ anchors the new continued-fraction theory. The exact order-40 data, the monotonicity conjecture, the endpoint-sensitive rate problem, and the regularized Mittag–Leffler problem provide concrete frontier targets suitable for both analysis and formalization.

# A Reproducibility instructions

From the directory containing this report, run

```
python3 rvachev_frontier_experiments.py --order 40 --digits 45
```

This regenerates:

- `up_even_moments.csv`;
- `up_jacobi_coefficients.csv`;
- `generated_tables.tex`;
- `numerical_checks.txt`;
- the two PDF/PNG figures.

The exact arithmetic uses Python's `fractions.Fraction`. The numerical checks use `mpmath`; plotting uses `matplotlib`. The script contains detailed comments, normalization statements, argument validation, and exact orthogonality checks.

# B Corpus audit ledger

The pinned 67-path inventory and the additional live draft families used in the audit are listed in the accompanying source fragment `corpus_inventory.tex`.

### Pinned 67-source ledger

| No. | Path relative to `Analysis/FabiusFunction/docs` |
|---|---|
| 1 | `FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex` |
| 2 | `Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex` |
| 3 | `Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex` |
| 4 | `fabius_lean_walkthrough/fabius_lean_walkthrough.tex` |
| 5 | `non-formalized-research-frontiers/non-formalized-research-frontiers.tex` |
| 6 | `archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex` |
| 7 | `archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex` |
| 8 | `archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex` |
| 9 | `archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex` |
| 10 | `archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex` |
| 11 | `archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex` |

| No. | Path relative to Analysis/FabiusFunction/docs |
|---|---|
| 12 | archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex |
| 13 | archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex |
| 14 | archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex |
| 15 | archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex |
| 16 | archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex |
| 17 | archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex |
| 18 | archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex |
| 19 | archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex |
| 20 | archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex |
| 21 | archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex |
| 22 | archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex |
| 23 | archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex |
| 24 | archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex |
| 25 | archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex |
| 26 | archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex |
| 27 | archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex |
| 28 | archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex |
| 29 | archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex |
| 30 | archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex |
| 31 | archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex |
| 32 | archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex |
| 33 | archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex |
| 34 | archive/research-frontiers/non-formalized-research-frontiers/non-formalized-research-frontiers.tex |
| 35 | archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex |
| 36 | archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex |
| 37 | archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex |
| 38 | non-formalized-research-frontiers/drafts/Fabius_Arithmetic_Rays_Frontier_Report/Fabius_Arithmetic_Rays_Frontier_Report.tex |

| No. | Path relative to `Analysis/FabiusFunction/docs` |
|---|---|
| 39 | `non-formalized-research-frontiers/drafts/Fabius_Half_Integer_Spectral_Frontier_Report/Fabius_Half_Integer_Spectral_Frontier_Report.tex` |
| 40 | `non-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report-2/Fabius_Rvachev_Frontier_Report.tex` |
| 41 | `non-formalized-research-frontiers/drafts/Fabius_Rvachev_Frontier_Report/Fabius_Rvachev_Frontier_Report.tex` |
| 42 | `non-formalized-research-frontiers/drafts/Fabius_Rvachev_New_Frontiers/Fabius_Rvachev_New_Frontiers.tex` |
| 43 | `non-formalized-research-frontiers/drafts/Fabius_Rvachev_Thue_Morse_Frontier_Results/Fabius_Rvachev_Thue_Morse_Frontier_Results.tex` |
| 44 | `non-formalized-research-frontiers/drafts/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy.tex` |
| 45 | `non-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex` |
| 46 | `non-formalized-research-frontiers/drafts/fabius_frontier_new_results/fabius_frontier_new_results.tex` |
| 47 | `non-formalized-research-frontiers/drafts/fabius_frontier_results_bundle/fabius_frontier_results.tex` |
| 48 | `non-formalized-research-frontiers/drafts/fabius_frontier_spectral_endpoint_report_bundle/fabius_frontier_spectral_endpoint_report.tex` |
| 49 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex` |
| 50 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex` |
| 51 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex` |
| 52 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex` |
| 53 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex` |
| 54 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex` |
| 55 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex` |
| 56 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex` |
| 57 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex` |
| 58 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex` |
| 59 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex` |
| 60 | `non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex` |
| 61 | `papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex` |
| 62 | `papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex` |
| 63 | `papers/arXiv-1502.06738v2/Lacunary_products.tex` |
| 64 | `papers/arXiv-1702.05442v1/09-Function.tex` |
| 65 | `papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex` |
| 66 | `papers/arXiv-2211.13570v1/document.tex` |

| No. | Path relative to `Analysis/FabiusFunction/docs` |
|---|---|
| 67 | `papers/ubiq15/ubiq15-corrected.tex` |

### Live additions checked after the pinned ledger

- `non-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/Fabius_Dyadic_Self_Sampling_Frontier_Report.tex`
- `non-formalized-research-frontiers/drafts/Fabius_Inverse_Frontier_Report_Source_and_PDF/Fabius_Inverse_Frontier_Report.tex`
- `non-formalized-research-frontiers/drafts/Fabius_Newton_Rvachev_Frontier_Report/Exponent_Sequence_Newton_Rvachev_Frontiers.tex`
- `non-formalized-research-frontiers/drafts/fabius_frontier_dyadic_inverse_barnes_report/` (report source and auxiliaries)

Additional auxiliary TeX fragments inside these live report packages were checked through their parent report structure. The pinned ledger remains the stable reproducible count; the live tree changes during active research and therefore should not be assigned a permanent count without a fresh commit-pinned traversal.

## C Selected exact data

The first eleven even moments and first twelve Jacobi coefficients are typeset in the main text. The CSV files contain the exact data through order 40. Selected late-order decimal values are:

| $n$ | $\beta_n$ | $1/4 - \beta_n$ | $n^2(1/4 - \beta_n)/(\log n)^2$ |
|---|---|---|---|
| 12 | 0.223692820061698 | 0.026307179938302 | 0.6135029154 |
| 15 | 0.229185734718218 | 0.020814265281782 | 0.6386018638 |
| 20 | 0.234873012373992 | 0.015126987626008 | 0.6742274757 |
| 25 | 0.238343193605764 | 0.011656806394236 | 0.7031551379 |
| 30 | 0.240650640426945 | 0.009349359573055 | 0.7273789024 |
| 35 | 0.242285352613889 | 0.007714647386111 | 0.7476320691 |
| 40 | 0.243489365492140 | 0.006510634507860 | 0.7655161747 |

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[1] J. Arias de Reyna, Arithmetic of the Fabius function, *Integers* **18** (2018), Paper A51.

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[5] V. Reshetnikov et al., *ProveIt: Analysis/FabiusFunction/docs*, GitHub repository, audited 27 August 2026, `https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs`.

[6] B. Simon, *Orthogonal Polynomials on the Real Line, Part 1: Classical Theory*, American Mathematical Society Colloquium Publications, vol. 54, AMS, 2005.

[7] W. Van Assche, The impact of Stieltjes' work on continued fractions and orthogonal polynomials, in *Thomas Jan Stieltjes: Collected Papers*, 1993; arXiv:math/9307220.